

Basics 32: Alfvén Waves from First Principles

V - Plasma waves and Alfvén physics

A full linear ideal-MHD derivation of the shear-Alfvén wave and its physical energy balance

Textbook draft, version 1.0. The shear-Alfvén wave is the central elementary wave of magnetized fluid plasma. Starting from a uniform equilibrium, this chapter derives its dispersion relation, polarization, wave equation, energy partition, and propagation along magnetic field lines. This chapter is designed for a reader with no prior plasma-physics training. It distinguishes exact identities from physical approximations and connects the central equations to later simulation modules.

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1 Learning objectives

After completing this note, the reader should be able to:

- Identify the equilibrium and perturbation assumptions of a shear-Alfvén wave.
- Derive linearized induction and momentum relations.
- Derive $\omega^2 = k_{\parallel}^2 v_A^2$.
- Interpret magnetic tension as the restoring force.
- Relate velocity and magnetic perturbation amplitudes.
- Derive kinetic-magnetic energy equipartition for a traveling wave.

2 Prerequisites and reading strategy

- Basics 24, 25, 29, and 31.

On a first reading, emphasize physical meaning and dimensional checks. On a second reading, reproduce the derivations. Attempt the computational laboratory only after the limiting cases are understood.

3 Uniform equilibrium and shear polarization

Take

$$\mathbf{B}_0 = B_0 \hat{\mathbf{z}}, \quad \rho_0 = \text{constant}, \quad p_0 = \text{constant}, \quad \mathbf{u}_0 = 0. \quad (3.1)$$

Consider incompressible transverse perturbations varying along the field:

$$\mathbf{u}_1 = u_x(z, t) \hat{\mathbf{x}}, \quad \mathbf{B}_1 = b_x(z, t) \hat{\mathbf{x}}. \quad (3.2)$$

Then $\nabla \cdot \mathbf{u}_1 = 0$, $\nabla \cdot \mathbf{B}_1 = 0$, and no density or pressure perturbation is required. The displacement is perpendicular to both $\mathbf{B}_0$ and the parallel wave vector.

4 Linearized induction

The ideal induction equation is

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{u} \times \mathbf{B}). \quad (4.1)$$

To first order,

$$\partial_t \mathbf{B}_1 = \nabla \times (\mathbf{u}_1 \times \mathbf{B}_0). \quad (4.2)$$

For the chosen geometry,

$$\boxed{\frac{\partial b_x}{\partial t} = B_0 \frac{\partial u_x}{\partial z}}. \quad (4.3)$$

A transverse velocity gradient bends the equilibrium field and creates a transverse magnetic perturbation.

5 Linearized momentum and magnetic tension

The transverse momentum equation is

$$\rho_0 \frac{\partial \mathbf{u}_1}{\partial t} = \frac{1}{\mu_0} (\nabla \times \mathbf{B}_1) \times \mathbf{B}_0. \quad (5.1)$$

For $\mathbf{B}_1 = b_x(z, t) \hat{\mathbf{x}}$,

$$\boxed{\rho_0 \frac{\partial u_x}{\partial t} = \frac{B_0}{\mu_0} \frac{\partial b_x}{\partial z}}. \quad (5.2)$$

This is magnetic tension: curvature of the perturbed field produces a restoring force along the transverse displacement.

6 Wave equation and Alfvén speed

Differentiate the momentum equation in time and use induction:

$$\rho_0 \frac{\partial^2 u_x}{\partial t^2} = \frac{B_0}{\mu_0} \frac{\partial}{\partial z} \left(\frac{\partial b_x}{\partial t} \right) \quad (6.1)$$

$$= \frac{B_0^2}{\mu_0} \frac{\partial^2 u_x}{\partial z^2}. \quad (6.2)$$

Therefore

$$\boxed{\frac{\partial^2 u_x}{\partial t^2} = v_A^2 \frac{\partial^2 u_x}{\partial z^2}}, \quad \boxed{v_A = \frac{B_0}{\sqrt{\mu_0 \rho_0}}}. \quad (6.3)$$

A plane wave $u_x = \tilde{u} e^{i(k_{\parallel} z - \omega t)}$ gives

$$\boxed{\omega^2 = k_{\parallel}^2 v_A^2}. \quad (6.4)$$

7 Amplitude and phase relation

For a plane wave, induction gives

$$-i\omega\tilde{b} = ik_{\parallel}B_0\tilde{u}. \quad (7.1)$$

Thus

$$\tilde{b} = -\frac{k_{\parallel}B_0}{\omega}\tilde{u} = \mp\sqrt{\mu_0\rho_0}\tilde{u}, \quad (7.2)$$

where the sign depends on propagation direction and phase convention. The velocity and magnetic perturbations are correlated; changing the propagation direction changes their relative sign.

8 Energy density and flux

The quadratic kinetic and magnetic perturbation energies are

$$\mathcal{E}_K = \frac{\rho_0 u_1^2}{2}, \quad \mathcal{E}_B = \frac{b_1^2}{2\mu_0}. \quad (8.1)$$

Using $|b_1| = \sqrt{\mu_0\rho_0}|u_1|$ for a traveling wave,

$$\boxed{\mathcal{E}_K = \mathcal{E}_B}. \quad (8.2)$$

Multiplying the momentum and induction equations by u_x and b_x/μ_0 yields

$$\frac{\partial}{\partial t} \left(\frac{\rho_0 u_x^2}{2} + \frac{b_x^2}{2\mu_0} \right) + \frac{\partial}{\partial z} \left(-\frac{B_0 u_x b_x}{\mu_0} \right) = 0. \quad (8.3)$$

The sign of the flux tracks propagation direction.

9 Elsasser variables

Define

$$\mathbf{z}^{\pm} = \mathbf{u}_1 \mp \frac{\mathbf{B}_1}{\sqrt{\mu_0\rho_0}}. \quad (9.1)$$

The linear equations become

$$\partial_t \mathbf{z}^{\pm} \pm v_A \partial_z \mathbf{z}^{\pm} = 0. \quad (9.2)$$

$\mathbf{z}^+$ and $\mathbf{z}^-$ describe disturbances propagating in opposite directions along the equilibrium field. This form is central in Alfvénic turbulence.

10 Limits of the elementary derivation

The result assumes uniform equilibrium, ideal conductivity, scalar pressure, no finite-Larmor-radius effects, and propagation exactly along $\mathbf{B}_0$. Nonuniform density creates an Alfvén continuum; toroidal geometry couples poloidal harmonics; kinetic particles add damping or drive. Those extensions motivate Basics 34-37 and Modules 4-6.

11 Computational laboratory

Integrate the coupled linear induction and momentum equations for a periodic domain. Initialize a single traveling wave and a standing wave. Verify the measured speed, amplitude relation, energy equipartition, and convergence of total perturbation energy in both Python and MATLAB.

12 Summary

- Shear-Alfvén waves are transverse ideal-MHD disturbances restored by magnetic tension.
- Their speed is $v_A = B_0/\sqrt{\mu_0\rho_0}$ and propagation is along the field.
- Velocity and magnetic perturbations have a direction-dependent phase relation.
- Traveling waves have equal kinetic and magnetic perturbation energies.
- Nonuniformity, geometry, and kinetic particles extend this elementary wave into continua and eigenmodes.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
v_A	Alfvén speed	$B_0/\sqrt{\mu_0\rho_0}$.
$k_{\parallel}$	parallel wave number	Phase variation along the equilibrium field.
u_x	transverse velocity	Shear-flow perturbation.
b_x	transverse magnetic perturbation	Field-line bending amplitude.
$\mathbf{z}^{\pm}$	Elsasser variables	Counterpropagating Alfvénic combinations.
$\mathcal{E}_K, \mathcal{E}_B$	perturbation energies	Kinetic and magnetic contributions.

13 Exercises

1. Derive the linear induction equation for the chosen polarization.
2. Derive the transverse Lorentz-force term from $(\nabla \times \mathbf{B}_1) \times \mathbf{B}_0$.
3. Obtain the Alfvén wave equation and check dimensions of v_A .
4. Derive the velocity-magnetic amplitude relation for both propagation directions.
5. Derive the quadratic energy conservation law.

14 References

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